

ARM-450 — Isometric Dimensioned Drawings

12 sheets · every printed part, the bought shaft, and the fit coupon

Pictorial isometric sheets in the conventional style: the part drawn in isometric with hidden lines removed, aligned linear dimensions with extension lines and arrow terminators, and leader callouts for every hole, radius, pocket and feature.

These are generated from the exported STL geometry by `draw_iso.py`, and every number is read from `cad/params.py`. A drawing here therefore cannot disagree with the part it describes — if a parameter changes, the drawing changes with it.

How to read these sheets

Projection. True isometric, viewed along (1, 1, 1). +Z is up the page, +X runs down to the right, +Y down to the left. All three axes are equally foreshortened, so a dimension measured along any axis is directly comparable to any other.

Hidden lines are removed, not dashed. Only edges actually visible from the viewpoint are drawn. Where you see a broken line it is a genuine feature edge, not a hidden one.

Colour carries meaning:

Dark navy	the part outline
Blue	linear dimensions — extension lines, dimension line, arrows both ends
Brown	leader callouts — holes, radii, pockets, notes

Units are millimetres throughout, to two decimals. Ø is a diameter, R a radius, DP a depth, THRU a through feature, BCD a bolt circle diameter.

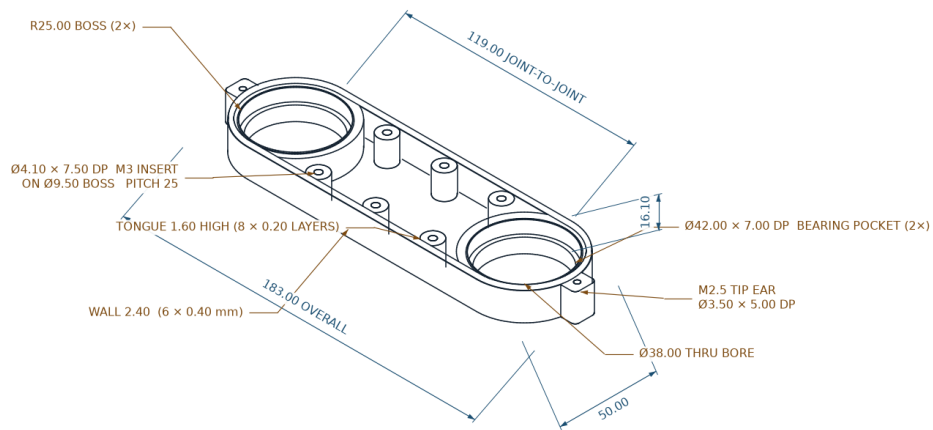
Tolerance, unless stated otherwise on the sheet: general ± 0.2 , bearing bores ± 0.05 . The 0.02 mm XY shrinkage measured on your printer is already allowed for in the modelled dimension — print to the number shown, do not compensate again.

Every bearing bore prints with its axis VERTICAL. Each title block repeats this. It is the single most consequential instruction on any of these sheets: your printer shrinks 0.02 mm in XY and nothing in Z, so a vertical bore comes out uniformly 0.02 mm under — which is the intended light press fit. Printed on its side the same bore shrinks in XY but not in Z and comes out oval, which reintroduces exactly the tilt and wobble the paired-bearing design exists to remove.

1. Link half — tongue side

LINK HALF — TONGUE SIDE

upper arm and forearm are identical parts



PART	link_half_tongue.stl
QTY	2 off (4 halves = 2 links)
MATERIAL	PLA+CF
MASS	47.9 g each
SECTION	50.0 x 29.0 stadium, tangent
ORIENT	split face DOWN on the bed, open side up
INFILL	10-15 % gyroid, 4 perimeters
TOL	bores ±0.05 · XY shrink 0.02 allowed for

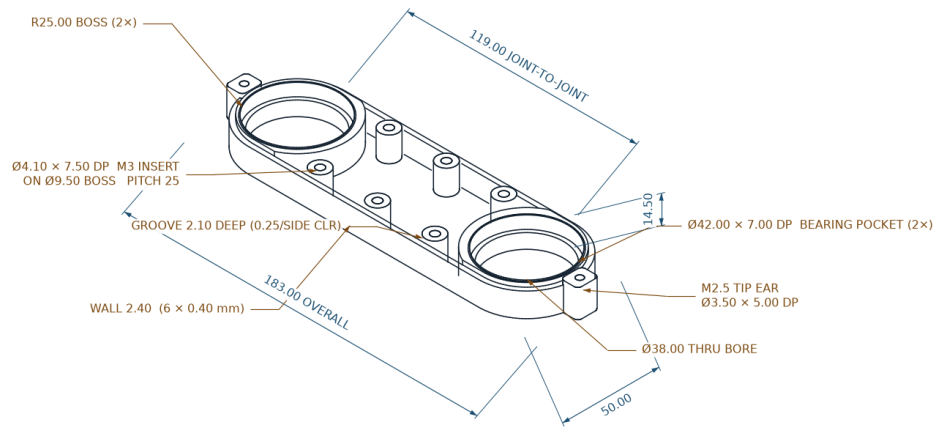
The two links are identical parts, so this drawing and the next cover all four halves. 119.00 joint-to-joint is the dimension that sets the kinematics — it appears in the URDF as L2 and L4_SPLIT, and generate_urdf.py asserts the two agree.

The section is a 50.0 x 29.0 stadium, and the R25.00 end boss is tangent to the straight flank. That tangency was the fix for the seam misalignment on your first print: with a half-height of 23.75 against a 25.0 boss radius the circle bulged past the rectangle and left a shallow 18.2° vertex, where any offset is amplified by $1/\sin \theta$ and 0.15 mm became 0.48 mm.

2. Link half — groove side

LINK HALF — GROOVE SIDE

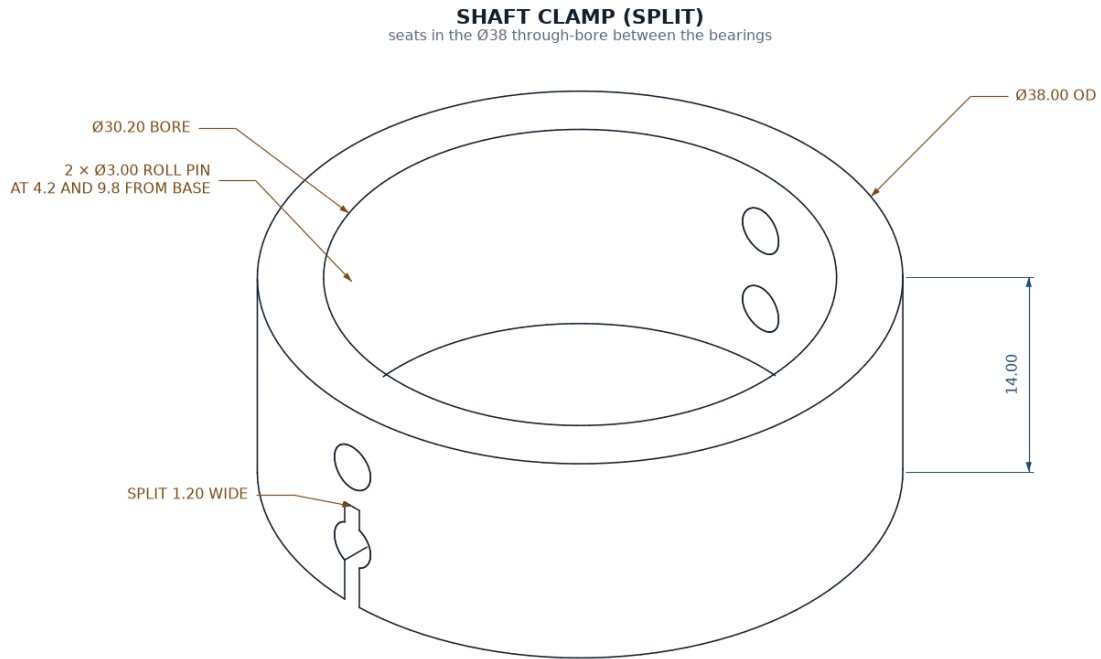
upper arm and forearm are identical parts



PART	link_half_groove.stl
QTY	2 off (4 halves = 2 links)
MATERIAL	PLA+CF
MASS	44.7 g each
SECTION	50.0 x 29.0 stadium, tangent
ORIENT	split face DOWN on the bed, open side up
INFILL	10-15 % gyroid, 4 perimeters
TOL	bores ±0.05 · XY shrink 0.02 allowed for

Identical to the tongue half except at the seam. The groove is cut 0.50 deeper than the tongue is tall, with 0.25 clearance per side. Both numbers exist because of the first print: a 1.5 mm tongue is 7.5 layers, which rounds to 1.60 and exactly equalled the groove depth, so the tongue bottomed out and held the halves apart. The tongue is now specified as $8 \times 0.20 = 1.60$, an integer number of layers by construction.

3. Shaft clamp



PART	shaft_clamp.stl
QTY	2 off
MATERIAL	PLA+CF
MASS	6,6 g each
ORIENT	bore axis VERTICAL
INFILL	100 %, 6 perimeters
NOTE	roll pin replaces grub screws – a grub on a 2 mm tube wall dents it

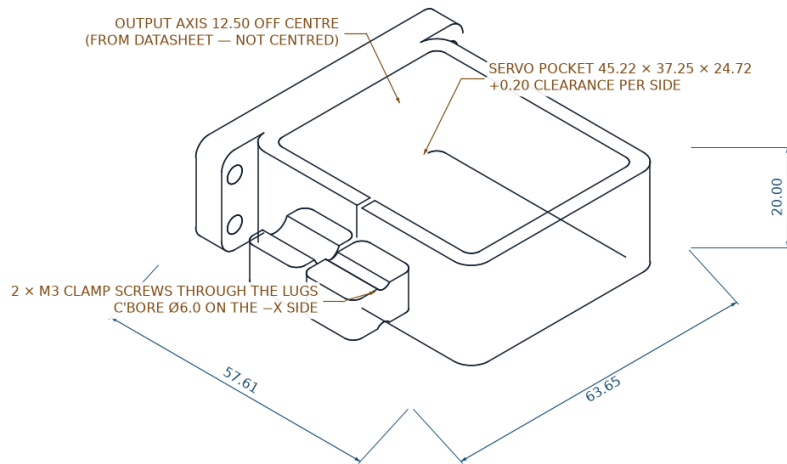
Seats in the Ø38.00 through-bore between the two bearings — not in the Ø42 pocket, which is only 7 mm deep and would leave 5 mm of the clamp hanging out.

Note the roll pin, not grub screws. The shaft is now a 2 mm wall aluminium tube, and an M3 grub tightened onto it puts about 170 MPa on the wall and simply dents it. A Ø3.00 roll pin works in double shear — 4241 N capacity against the 196 N the joint torque demands.

4. Servo collar

SERVO COLLAR

full-perimeter clamp — closed loop, not a cantilever



PART	servo_collar.stl
QTY	2 off
MATERIAL	PLA+CF
MASS	22.3 g each
ORIENT	collar axis VERTICAL
INFILL	100 %, 6 perimeters
NOTE	carries stall torque as shear flow around a closed loop: 0.25 MPa

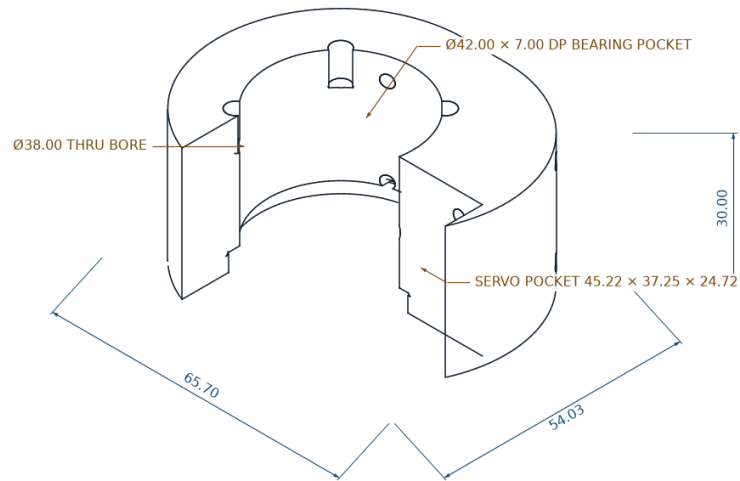
A full-perimeter clamp: a closed loop, not a cantilever bracket. This is the part that replaced the mount that snapped. Sized against the servo's stall torque rather than its gravity load, it carries that torque as shear flow around a closed section at 0.25 MPa, against the 94 MPa the original cantilevered stub saw.

The servo output axis is 12.50 off centre — from the datasheet, not assumed. Every pocket in this design was originally drawn centred, which was wrong.

5. J4 roll housing

J4 ROLL HOUSING

forearm roll — carries the J4 servo

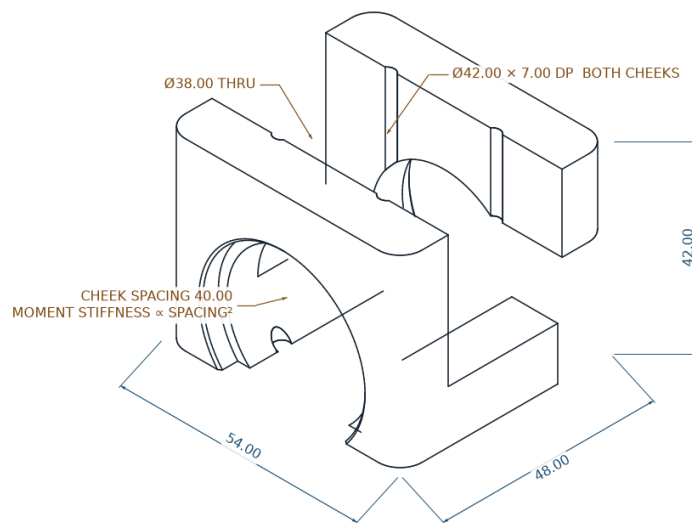


PART	wrist_j4_housing.stl
QTY	1 off
MATERIAL	PLA+CF
MASS	47.3 g
ORIENT	bearing bore axis VERTICAL
INFILL	50-60 % gyroid, 6 perimeters
TOL	bearing bore ±0.05

6. J5 pitch yoke

J5 PITCH YOKE

the fork that carries the wrist pitch axis



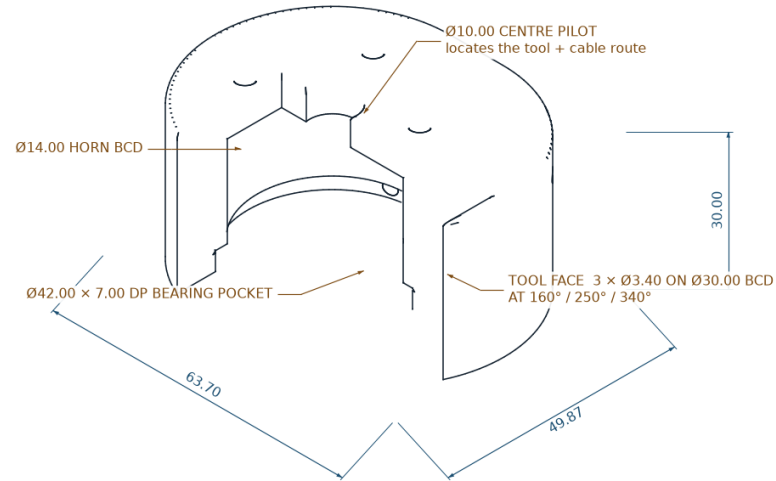
PART	wrist_j5_yoke.stl
QTY	1 off
MATERIAL	PLA+CF
MASS	40.1 g
ORIENT	bearing bore axis VERTICAL
INFILL	50-60 % gyroid, 6 perimeters
TOL	bearing bore ± 0.05

Cheek spacing 40.00 is a stiffness dimension, not a packaging one. Moment stiffness goes as the square of bearing spacing, so this single number is worth more than any amount of extra wall thickness. It is what takes joint wobble from 28.9 mm at the tool down to 0.65 mm.

7. J6 output / tool flange

J6 OUTPUT / TOOL FLANGE

tool roll — the pen or gripper bolts here



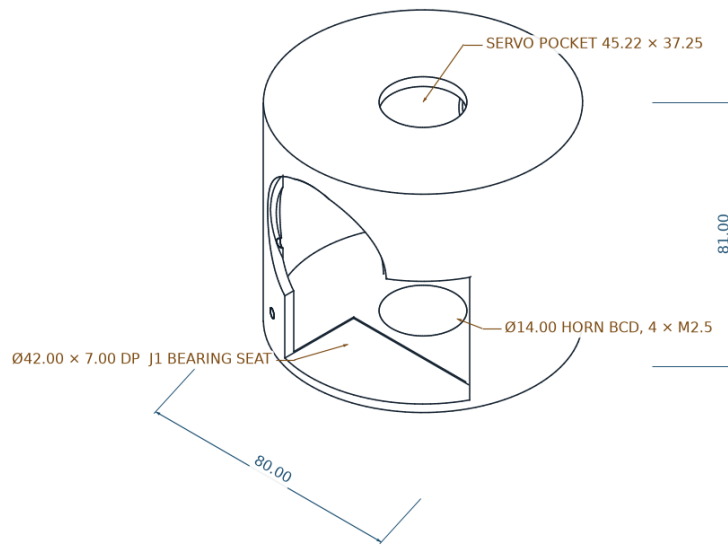
PART	wrist_j6_output.stl
QTY	1 off
MATERIAL	PLA+CF
MASS	42.4 g
ORIENT	bearing bore axis VERTICAL
INFILL	50-60 % gyroid, 6 perimeters
TOL	bearing bore ± 0.05

The tool face is the standard mounting interface: 4 $\times$ M3 on a $\varnothing 30.00$ bolt circle with a $\varnothing 10.00$ centre pilot. The pen holder for the sine trace, and later the MASCOT docking hardware, both bolt here.

8. J1 turret

J1 TURRET

carries the shoulder; rotates on the base bearing

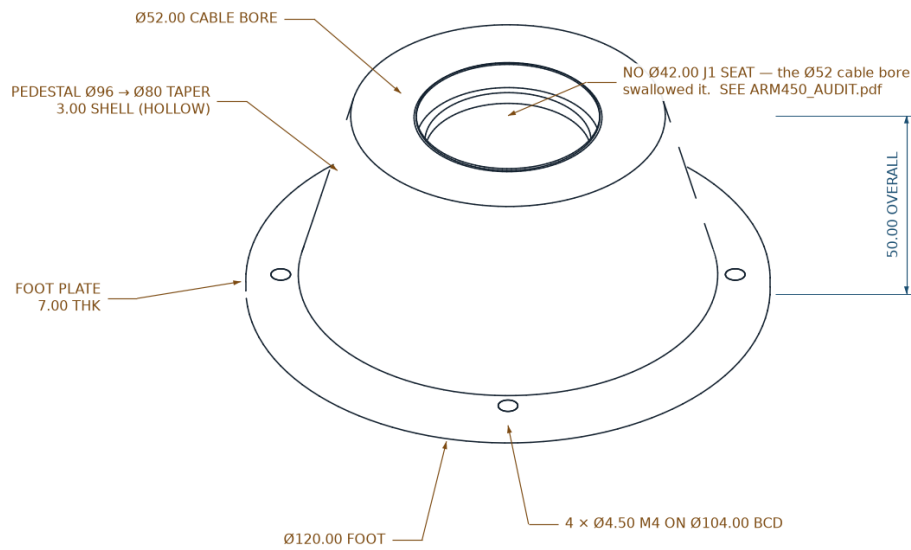


PART	turret_j1.stl
QTY	1 off
MATERIAL	PLA+CF
MASS	73.5 g
ORIENT	bearing seat DOWN, axis vertical
INFILL	50-60 % gyroid, 6 perimeters

9. Base pedestal

BASE PEDESTAL

hollow — solid it was 166 g, 14 % of the whole budget

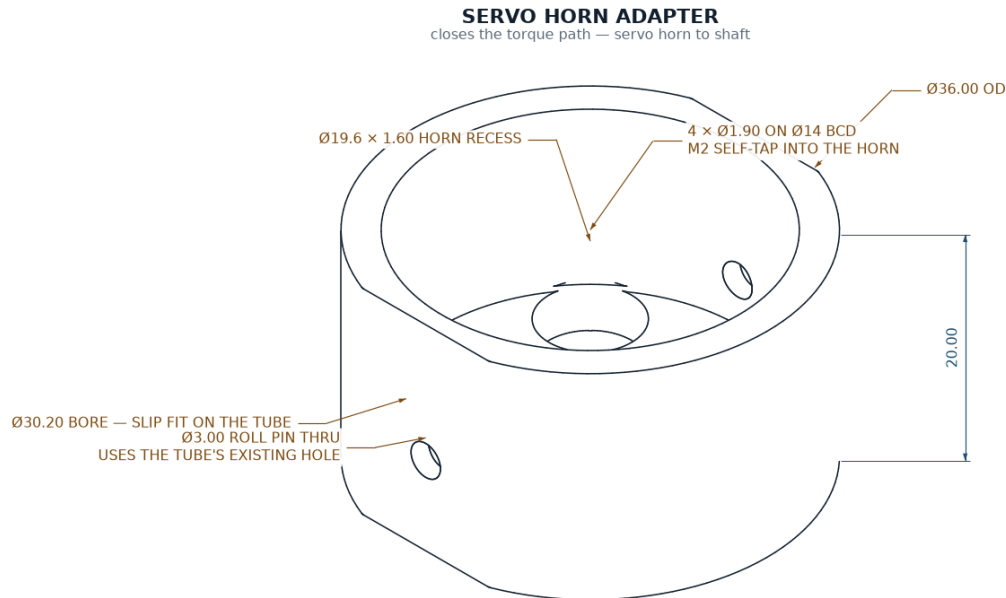


PART	base.stl
QTY	1 off
MATERIAL	PLA+CF
MASS	127.6 g
ORIENT	foot DOWN, upright
INFILL	50-60 % gyroid, 6 perimeters
NOTE	longest print at ~5.6 h — run overnight

Hollow, with a 3.00 shell. Modelled solid this part was $234 \text{ cm}^3 = 166 \text{ g}$ — the single largest printed item and 14 % of the whole mass budget for something that just sits on the table. The shell keeps the same outside and the same bearing seat.

At ~5.6 h it is also the longest single print, and the least critical to joint fit, which is why it goes last and runs overnight.

10. Servo horn adapter



PART	horn_adapter.stl
QTY	6 off — one per joint
MATERIAL	PLA+CF
MASS	8.2 g each
ORIENT	horn face DOWN on the bed
INFILL	100 %, 6 perimeters
SIZED ON	servo STALL 2.94 N·m: horn screws 105 N, SF 4.6 roll pin 196 N, SF 7.3 reuses the pin hole the tube already has 12 mm from each end
NOTE	

The part that closes the torque path. Servo horn → adapter → roll pin → tube → clamp → link. Before it existed there was none, and nothing in the CAD flagged that: the old solid joint_shaft carried a Ø20 bolt circle on its own shoulder, and when the shaft became a bought tube the shoulder went with it. Separately, horn_pattern() was being called on three wrist parts but cutting at $r = 7.0$ — inside the Ø38 bore, so a silent no-op. The geometry audit found both at once.

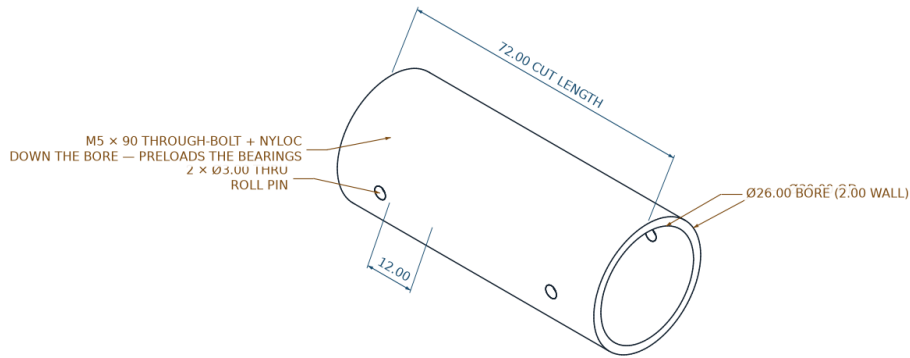
It reuses the roll-pin hole the tube already has 12 mm from each end, so the tube needs no extra machining: one pin drives the clamp at one end, one drives this at the other.

Sized against servo stall, 2.94 N·m — horn screws 105 N each (bearing 13.1 MPa, SF 4.6, which is why the face is 4 mm and not thinner), roll pin 196 N (SF 7.3), hub torsion SF 40.

11. Joint shaft — aluminium tube

JOINT SHAFT — ALUMINIUM TUBE

BOUGHT PART: cut and drill only, do not print



PART	shaft_tube.stl (bought stock)
QTY	6 off
MATERIAL	AL6061 / 6060 tube
STOCK	Ø30 x 2 wall, ~500 mm
MASS	34.2 g each · 205 g total
CHECK	MEASURE OD before cutting: 29.98–30.01 use as-is to 29.92 Loctite 603/638 below turn or buy drawn tube
NOTE	printing fails on the preload thread and press-fit creep, not strength

This is a bought part. Cut to length and drill two cross holes; do not print it.

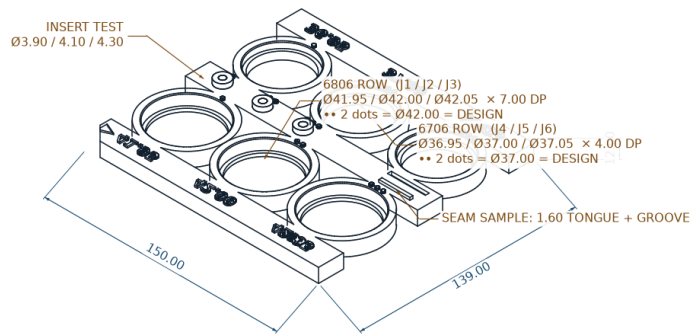
Printing is not excluded by strength — at servo stall a printed Ø30 shaft sees 0.55 MPa shear (SF 32) and 0.57 MPa bending (SF 104). It is excluded by two local effects: the M5 preload thread strips, and the press fit creeps away as the plastic relaxes. So metal goes only where metal is needed — the tube provides the bearing seats, a steel M5 through-bolt carries the preload down the Ø26.00 bore, and a roll pin handles torque.

Measure the OD before cutting. Extruded tube is held to about ± 0.10 – 0.20 on OD; a bearing seat wants roughly $+0.011/+0.002$. That is 10–20× looser than needed, and the title block on the sheet gives the decision table.

12. Fit-test coupon

FIT-TEST COUPON

PRINT THIS FIRST — it gates every part with a bearing pocket



PART	fit_coupon.stl
QTY	1 off
MATERIAL	same filament as the parts it gates
MASS	69.6 g · -2.3 h
ORIENT	flat, pockets facing UP
ROWS	top Ø42 = 6806, J1/J2/J3 bottom Ø37 = 6706, J4/J5/J6
PURPOSE	2.3 h spent here protects the 19 h of parts that carry a bearing pocket
READ IT	firm thumb press and sits square = right drops in free = too big will not start = too small

Print this first. It carries three bearing pockets (Ø41.95 / Ø42.00 / Ø42.05), three insert-boss hole sizes and a seam sample, in 27 g and about 0.9 h.

It is a hard gate. If Ø42.00 turns out wrong, BRG_FIT changes and every part with a bearing pocket is scrapped — roughly 19 of the 21 total print hours. Spending 0.9 h to protect 19 h is the entire argument for it.

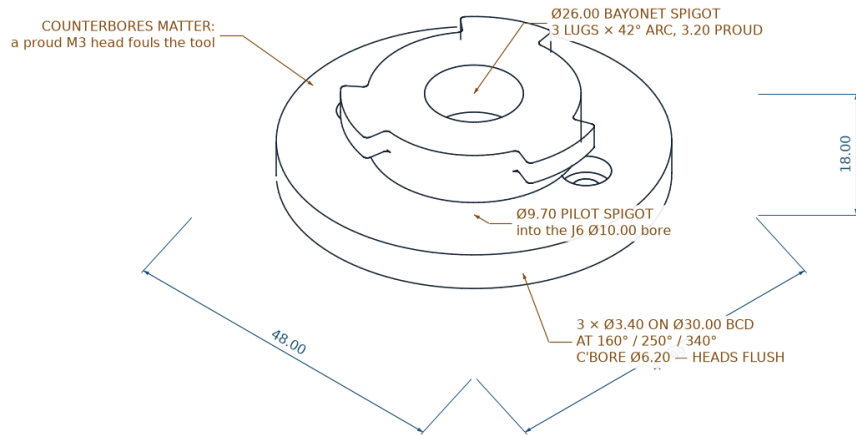
End effectors

The six quick-change parts. Full description, the coupling concept and the docking sequence are in ARM450_END_EFFECTORS.pdf; these are the shop sheets.

13. Tool adapter — the arm side

TOOL ADAPTER — ARM SIDE

bolts to J6 ONCE and stays there; every tool bayonets onto this



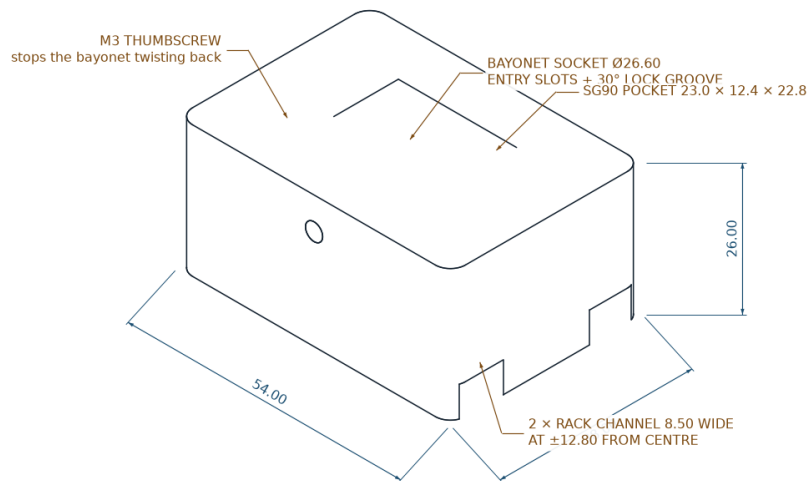
PART	tool_adapter.stl
QTY	1 off
MATERIAL	PLA+CF
ORIENT	spigot UP, flat face on the bed
INFILL	60 %, 6 perimeters
TOL	lug thickness ±0.10 — it sets the bayonet preload
MASS	14.4 g at 90 % fill

Bolts to J6 once and stays there. Three M3 on the Ø30.00 circle at 160 / 250 / 340° — the same three the J6 face has, because the servo pocket eats the fourth quadrant. The counterbores are not cosmetic: a proud M3 head fouls the tool sliding over it.

14. Gripper body

GRIPPER BODY

parallel jaws, rack and pinion, one SG90



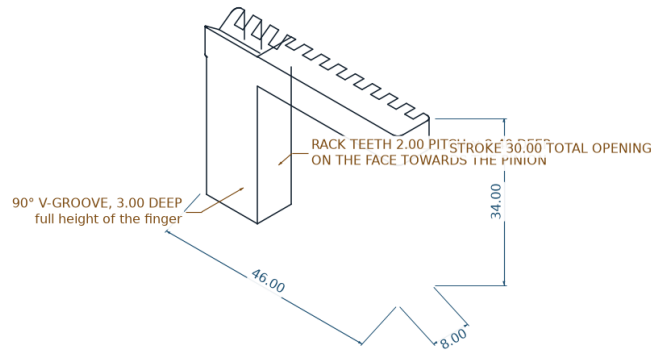
PART	tool_gripper.stl
QTY	1 off
MATERIAL	PLA+CF
ORIENT	socket face UP
INFILL	40 %, 4 perimeters
NOTE	RACK_OFFSET is NOT PCD/2 + JAW_T/2: the teeth are cut 2.40 into the bar, so the pitch line sits that much in
MASS	43.4 g at 90 % fill

One SG90 in a $23.0 \times 12.4 \times 22.8$ pocket, driving a $\varnothing 20$ pinion between two rack channels at ± 12.80 from the centre. That offset is not simply $PCD/2 + JAW_T/2$: the teeth are cut 2.40 into the bar, so the pitch line sits that much inside the bar face.

15. Gripper jaw

GRIPPER JAW

print two — mirror the second in the slicer



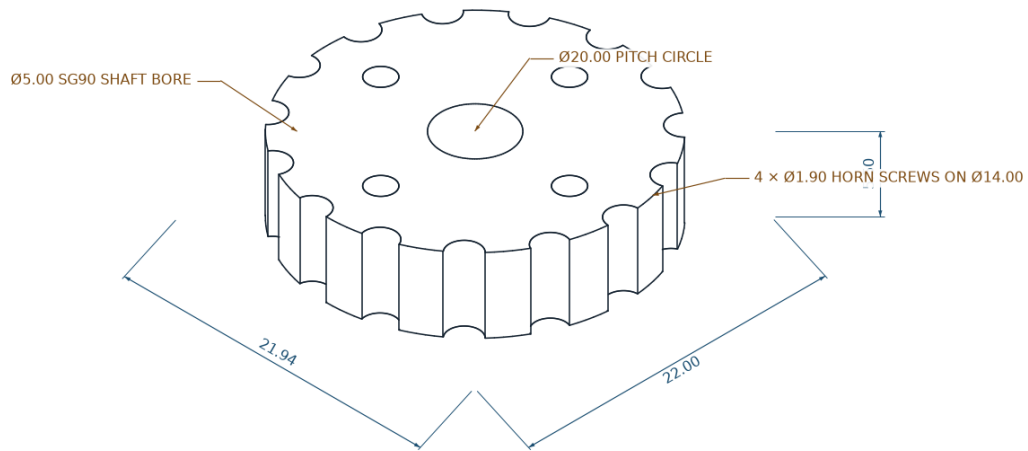
PART	gripper_jaw.stl
QTY	2 off (one mirrored)
MATERIAL	PLA+CF
ORIENT	rack bar FLAT on the bed, teeth sideways
INFILL	100 % — 2.00 teeth print hollow otherwise
WHY THE V	a flat jaw touches a cylinder on ONE line and it rolls out under load; a 90° V touches on four
MASS	5.7 g at 100 % fill

Print two, the second mirrored in the slicer. 100 % infill — the teeth are 2 mm and sparse infill leaves them hollow. The 90° V in the gripping face is what stops a cylinder rolling out sideways under load: a flat jaw touches a round part on one line, a V touches on two per jaw and locates it on its own axis whatever the diameter.

16. Gripper pinion

GRIPPER PINION

one turn of the SG90 drives both racks together



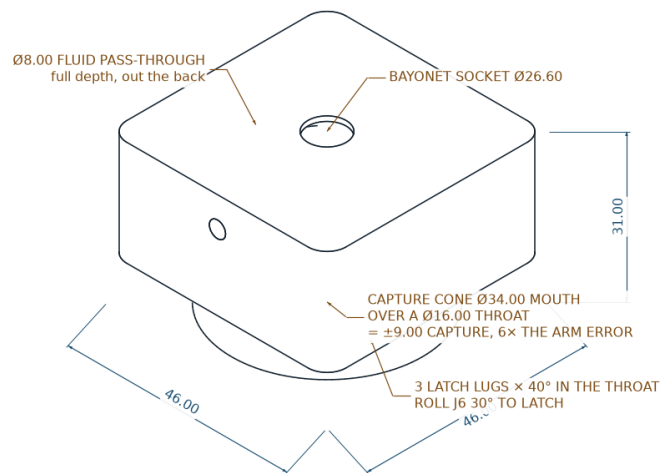
PART	gripper_pinion.stl
QTY	1 off
MATERIAL	PLA+CF
ORIENT	flat, teeth in the XY plane
INFILL	100 % - small teeth, high tooth load
NOTE	PCD raised 16 → 20 on 2026-08-22: at 16 the rack pitch line sat 1.20 outside the PCD and never touched
MASS	2.3 g at 100 % fill

Ø20 pitch circle, Ø5 SG90 shaft bore, four Ø1.90 horn screws on Ø14. 100 % infill.

17. MASCOT docking probe

MASCOT DOCKING PROBE

latched by J6 itself — no seventh actuator



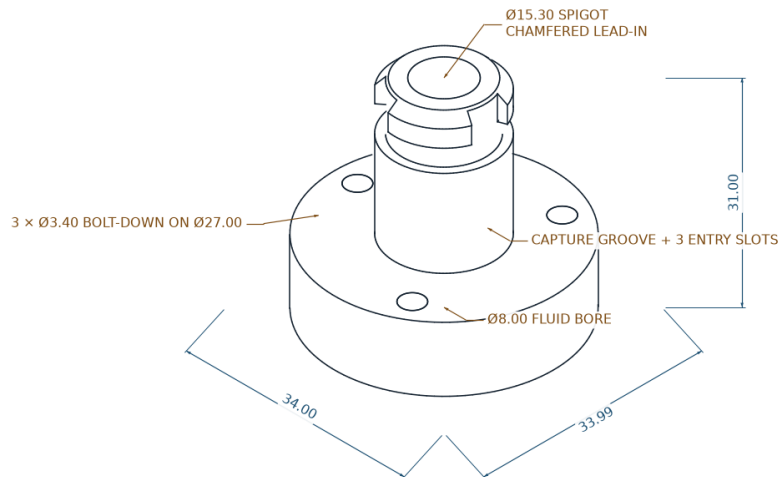
PART	tool_dock.stl
QTY	1 off
MATERIAL	PLA+CF
ORIENT	cone DOWN on the bed, socket up
INFILL	40 %, 4 perimeters
NOTE	the cone is the whole point: it turns a ±1.4 mm arm into a ±9 mm target
MASS	48.0 g at 90 % fill

Ø34 mouth funnelling to a Ø16 throat, with three latch lugs standing inward from the throat wall. No seventh actuator — J6 is the tool roll axis and rolling it 30° turns the latch.

18. Docking target port

DOCKING TARGET PORT

the passive side — bolt it to the bench, no arm needed



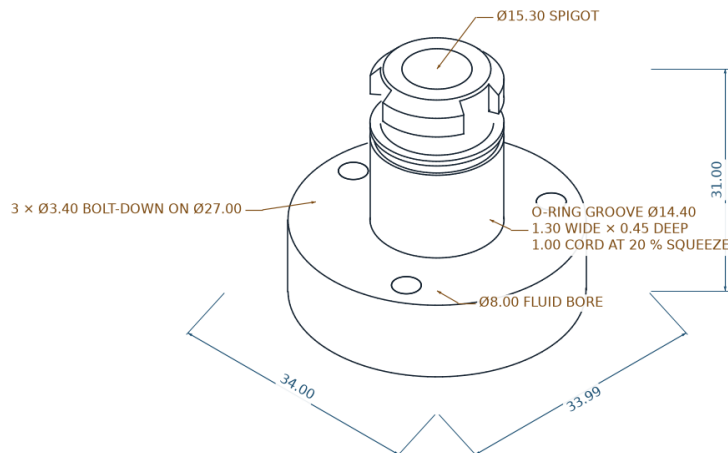
PART	dock_target.stl
QTY	1 off
MATERIAL	PLA+CF
ORIENT	flat face on the bed, spigot up
INFILL	40 %, 4 perimeters
NOT ARM	bench fixture — NOT in the 1450 g budget
PRINT IT	FIRST of the tools: it proves the docking interface with no arm, no fuel, no flight software
MASS	12.6 g at 90 % fill

The passive half, bolted to the bench. A 21 mm spigot — deliberately longer than the 9 mm cone is deep, so the post reaches the latch before the cone can ground on the flange — with the capture groove cut 1.20 mm into its outer face and three entry slots for the lugs.

19. Docking target port — SEALED

DOCKING TARGET PORT — SEALED

the same port with an O-ring groove, for fluid transfer



PART	dock_target_sealed.stl
QTY	1 off — ALTERNATIVE to dock_target
MATERIAL	PLA+CF
ORIENT	flat face on the bed, SPIGOT UP
INFILL	40 %, 4 perimeters
WHY UP	the groove must be a HORIZONTAL feature. Printed on its side it is a stack of stair-steps and the O-ring cannot seal radial, not a face seal — the bayonet has 0.5 mm of axial play at each end
SEAL	
MASS	12.5 g at 90 % fill

The same port with a $\text{Ø}14.40 \times 1.30 \times 0.45$ deep O-ring groove in the spigot, for fluid transfer. An alternative to dock_target, not an addition — print whichever the demonstration needs.

The seal is radial, not a face seal, and that is not a stylistic choice: the bayonet has 0.5 mm of axial play at each end of the capture groove, so a face seal would be fully loaded at one extreme of that play and completely unloaded at the other. A radial seal's squeeze is set by diameters and does not care where the spigot sits.

Print spigot UP. The groove has to be a horizontal feature; on its side it becomes a stack of stair-steps that an O-ring cannot seal against.

Regenerating these sheets

```
cd ~/ros2_ws/arm450_design
python3 draw_iso.py # rewrites figures/iso/*.png
python3 build_pdf.py DRAWINGS_ISO.md
```

Change anything in cad/params.py, rebuild the CAD, and re-run — the dimensions follow automatically. There are no hand-typed numbers on these sheets to fall out of date.